

Language Models for Structured Data Analysis



Presented by

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PROBLEM STATEMENT

- Most people cannot query a database; SQL is a technical barrier between everyday users and the data they need.
- Problem: A sales manager wants to know which products are underperforming. They need a data analyst, a ticket, and days of waiting just to answer **one** question.
- What we're building: System where that same manager types a question and gets answer instantly with no **SQL** required

DATASET: Spider 1.0 Yale text-to-SQL benchmark

	Type	Schema
Tables (14)		
> Asset_Parts		CREATE TABLE `Asset_Parts`
> Assets		CREATE TABLE `Assets` (`a:
> Engineer_Skills		CREATE TABLE `Engineer_Ski
> Engineer_Visits		CREATE TABLE `Engineer_Vis
> Fault_Log		CREATE TABLE `Fault_Log` (
> Fault_Log_Parts		CREATE TABLE `Fault_Log_Pa
> Maintenance_Contracts		CREATE TABLE `Maintenance
> Maintenance_Engineers		CREATE TABLE `Maintenance
> Part_Faults		CREATE TABLE `Part_Faults`
> Parts		CREATE TABLE `Parts` (`par
> Skills		CREATE TABLE `Skills` (`skil
> Skills_Required_To_Fix		CREA
> Staff		
> Third_Party_Companies		

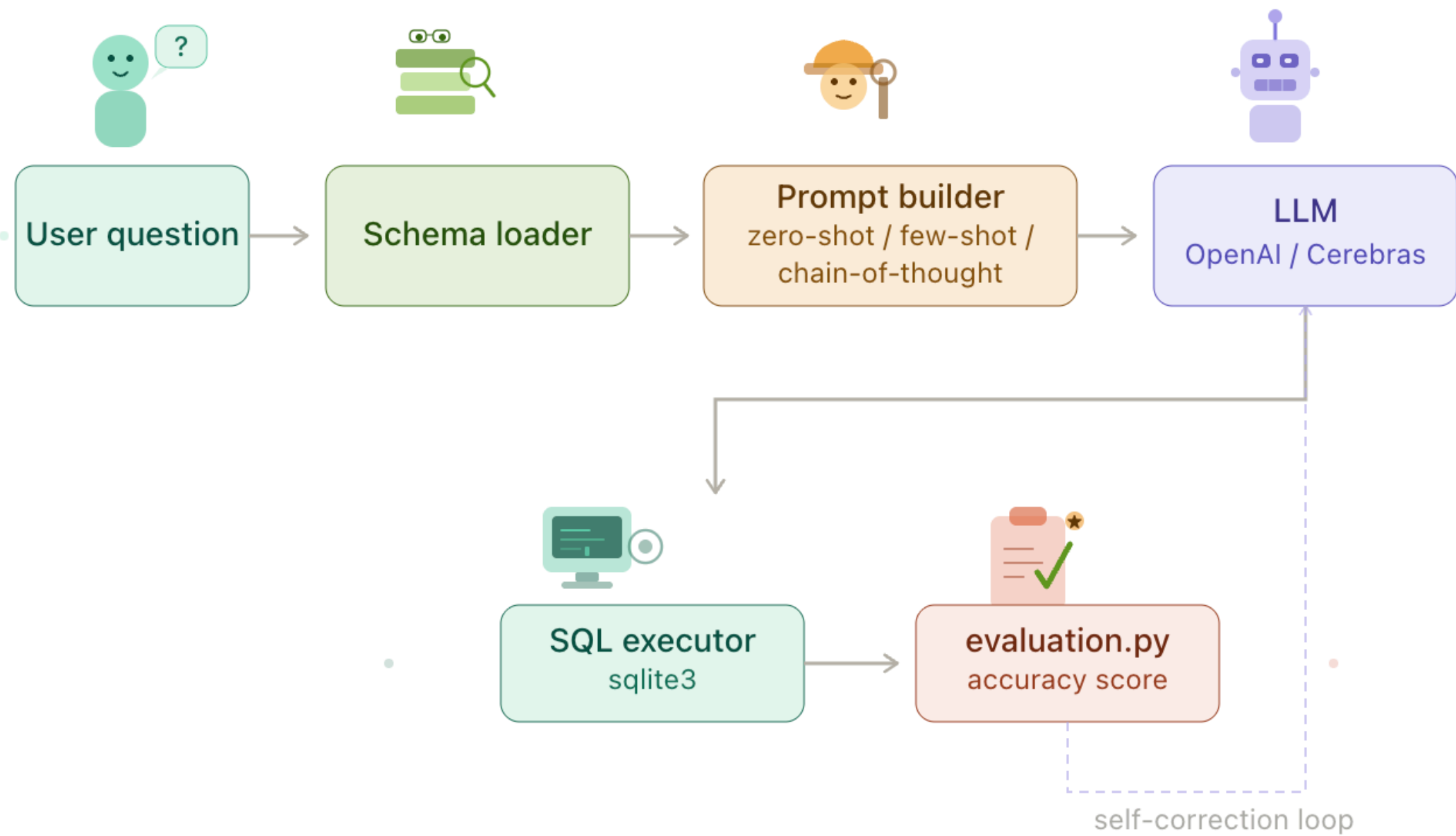
- **What makes it hard**
 - Questions span domains the model has never seen during training
 - Requires real SQL joins, subqueries, and aggregations (not simple lookups)
- **Test setup**
 - Using the official test split: [test.json](#) for questions, [test_tables.json](#) for schemas, [test_database/](#) for execution, [test_gold.sql](#) as answer key



Question pairs
10,181
Natural language → SQL

Unique databases
200
Across 138 domains

Unique SQL queries
5,693
Complex, multi-table



End-to-end Pipeline Architecture

Implemented (progress)

1. Zero-shot, Self-correction, Few-shot, CoT
2. RAG pipeline:
 - a. Lexical RAG: retrieves by exact keyword overlap
 - b. Hybrid RAG: combines keyword matching with semantic embeddings to handle synonyms and renamed schema terms.

Methodology

1. Used Spider 1.0 for cross-domain text-to-SQL evaluation.
2. Build pipeline: question -> schema retrieval -> SQL generation -> execution -> result.
3. Tested prompting strategies
4. Compared lexical and hybrid RAG for schema retrieval
5. Measured execution accuracy, query accuracy, retrieval quality, and robustness
6. Used ablation studies to compare what improved performance.

1) Multi-dimensional ablation review (Pending)

Controlled cost analysis on full length of retrieval strategies × prompting strategies × model size through precise removal of parts

2) Hybrid-Lexical RAG

Ensures robust retrieval by analyzing queries conceptually AND with exact keyword matching

3) Agentic Intent Disambiguation & Self Correction (Tentative)

Clarifying ambiguous user request intent with interactive dialogue, fixes erroneous SQL statements in cases of hallucinations or syntactic errors

4) Sovereign Masking (Pending)

Privacy abstraction layer to conceal schema names from LLMs in case of sensitive data

Baselines, Metrics, & Setup

Evaluation Metrics

- Main Metric: Execution accuracy
- Improve RAG:
 - Schema Retrieval Recall@k
 - Strict Recall Rate
- Dr. Spider (Robustness)

Future:

- Test Suite Accuracy (side metric)
- Valid Efficiency Score

Experimental Setup

1. 320 questions from test split, stratified across all difficulty levels
2. Model: [Cerebras Llama 3.1 8B](#) (free tier, reproducible), [OpenAI GPT-5.4](#)
3. Schema sourced from [test_tables.json](#), executed on [test_database/](#)
4. Gold answers sliced from [test_gold.sql](#)
5. Scored with Spider's official [evaluation.py](#)
6. Prompting strategies: zero-shot, chain-of-thought, few-shot, self-correction
7. Retrieval settings: no RAG, lexical RAG, hybrid RAG

Baseline

Cerebras Llama 3.1 8B

Strategy	Easy	Medium	Hard	Extra	Overall
Zero-shot	60.0%	56.2%	42.5%	35.0%	48.4%
Chain-of-thought	70.0%	32.5%	30.0%	10.0%	35.6%
Few-shot	72.5%	75.0%	45.0%	48.7%	60.3%
Self-correction	65.0%	57.5%	47.5%	36.2%	51.6%

Few shot GPT
accuracy
72.5%
strongest strategy

GPT 5.4

Strategy	Easy	Medium	Hard	Extra	All
Zero shot	60.0%	63.7%	53.7%	21.2%	49.7%
Chain of thought	50.0%	62.5%	60.0%	22.5%	48.7%
Few shot	97.5%	78.7%	76.2%	37.5%	72.5%
Self correction	58.8%	58.8%	52.5%	18.8%	47.2%

- **Easy:** simple queries, usually one table and minimal SQL machinery.
- **Medium:** a bit more structure, like filters, grouping, ordering, or a simple join.
- **Hard:** more complex SQL, such as multiple joins, multiple aggregations, or a nested query.
- **Extra:** the most structurally complex queries, usually combining several hard features at once.

Key Insights

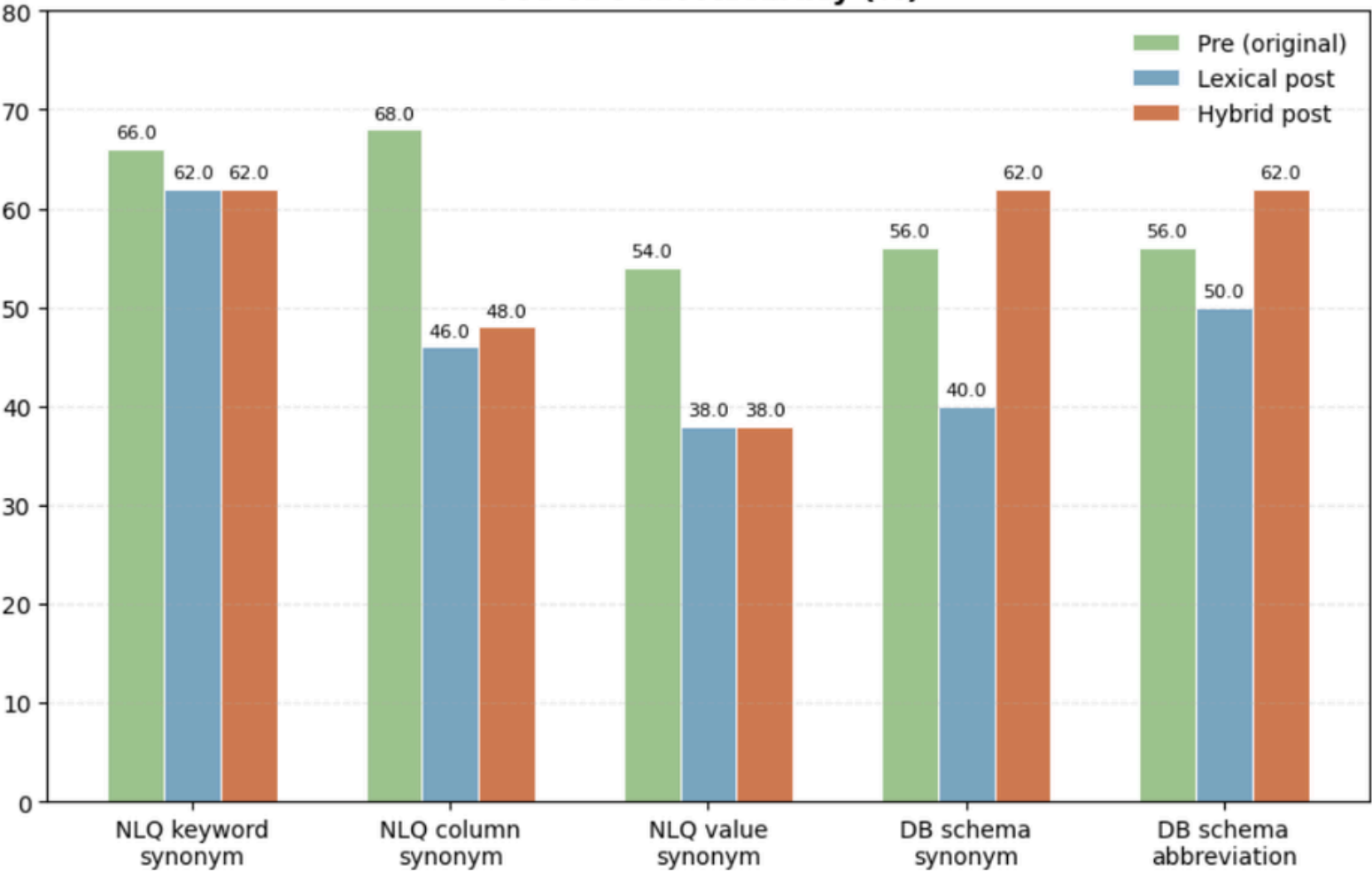
Setup	Recall@k	Precision@k	SRR	Avg Tables Returned
Old lexical (top-4)	100.0%	43.5%	100.0%	3.70
New lexical (top-3 + score-ratio)	97.7%	61.6%	95.0%	2.61
Hybrid (top-3)	97.7%	59.8%	95.0%	2.79

Model	Strategy	Easy	Medium	Hard	Extra	All
Cerebras	Few-shot	72.5%	75.0%	45.0%	48.7%	60.3%
Cerebras	Lexical + Self-Corre ction	72.5%	78.8%	57.5%	45.0%	63.4%

Strategy	Extra Exec Accuracy
Current RAG (top k=3)	52.5%
Broader RAG (top k=4)	56.2%
Full schema	60.0%

Dr.Spider: Lexical vs Hybrid Robustness

Pre vs Post Accuracy (%)



- Initially our RAG had very similar results to OG few shot → looked into ways to improve RAG using Recall@k/Precision@k
- Lexical + Self Correction helped improve Medium, Hard, but Extra still struggles
- Used full schema for Extra only and accuracy jumped (more schema context aids harder queries but is too much noise for easy/medium)
- Lexical good for simple direct match cases
- Hybrid specifically good for DB name changes (which can aid us with sovereign masking)



TECHNICAL ROADMAP:

- Streamlit demo
- live end-to-end query -> SQL
-> result app
- exact-match/query-
accuracy reporting
- Full ablation table: every
strategy × with/without an
RAG × accuracy + cost
- Improve RAG
- Sovereign Masking
(MaskSQL)

-
- Agentic Intent
Disambiguation
 - System Prompt Engineering
for involved LLMs



EXPECTED CONTRIBUTIONS

- **For researchers:** A controlled ablation showing which prompting strategies/RAG techniques matter and by how much — not just "it works" but "here's the cost-benefit."
- **For builders:** A reproducible open-source pipeline using free models — showing what's achievable without GPT level costs.

NEXT STEPS AND EXPECTED CONTRIBUTIONS



THANK YOU!

